

Coen Adler

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Office: DBH 4059

Location: Irvine, CA

Research interests

Mechanistic Interpretability, eXplainable AI (XAI), Time Series and Tabular Foundation Models

Education

University of California, Irvine

Irvine, California

PhD in Computer Science

Sep 2024 – June 2029

Advisor: Professor Padhraic Smyth.

GPA: 3.97

University of California, Santa Cruz

Santa Cruz, California

BS in Computer Science

Sep 2021 – June 2024

Advisors: Professors Leilani Gilpin, Luca de Alfaro.

GPA: 3.94

Papers

Calibration Properties of Time Series Foundation Models: An Empirical Analysis

Coen Adler, Yuxin Chang, Samar Abdi, Padhraic Smyth

ICML Workshop on Foundation Models for Structured Data, 2025.

Model Pruning and Explainability: An Experience Report

Coen Adler, Leilani Gilpin, Tyler Sorenson

UC Santa Cruz - Undergraduate Thesis, 2024.

Diffusion Models for Species-Specific Functional Habitat Connectivity

Natalia Ocampo-Peñuela, Luca de Alfaro, Coen Adler, et al.

In Progress, 2023.

Fair Augmentation of Decision Trees Through Selective Node Retraining

Coen Adler, Leilani Gilpin.

BayLearn Symposium - Abstract, 2023.

Utilizing Segment Anything Model for Assessing Localization of GRAD-CAM in Medical Imaging

Evan Kellener, Ihina Nath, An Ngo, Thomas Nguyen, Coen Adler, et al.

arXiv, 2023.

Analysis of Training PINNs on Multiple Computational Fluid Dynamics Models Consisting of Different Shapes

Bryce Montgomery, Kevin Bachelor, Manju Shettar, Coen Adler, et al.

Unpublished Manuscript, 2023.

Research Labs

The Datalab

Advisor: Padhraic Smyth (UCI)

September 2024 – Present

Broad AI/ML research lab with current research interests in algorithmic fairness and robust ML systems, human-AI interaction, and deep learning for predictive modeling.

ACM Undergraduate AI Research Lab

Advisors: **Coen Adler**, Arnav Kartikeya (UCSC)

March 2023 – June 2024

Undergraduate AI-focused research lab with the goal of introducing AI research to passionate undergraduate students: Usage of Meta's Segment Anything Model on GRAD-CAM for medical segmentation, Applications of Physics-Informed Neural Networks on complex geometric aerodynamics.

Artificial Intelligence Accountability Explainability Lab

Advisor: Professor Leilani Gilpin (UCSC)

December 2021 – June 2024

Explainable AI with work in robust and fair AI systems: GAN applications in AV robustness, Benchmarking disagreements of SHAP and LIME explanations, Fair augmentation of Decision Trees. Led lab organization and resource/computing usage.

Computational Ecology Lab

Advisor: Professor Luca de Alfaro (UCSC)

December 2021 – June 2024

Applications of game theory and machine learning to the study of interactions: Diffusion model for mapping forest bird habitat connectivity, Architecture for bird mapping browser tool.

Research Projects

Analyzing Post-hoc Explanation Disagreements

Sep 2022 – Dec 2022

Quantifying disagreements between post-hoc explanations: SHAP and LIME explanations often show minimal correlation, disagreements measured through a combination of cosine similarity and linear residuals on a per feature comparison.

GANs in Adverse AV Vision Feature Generation

Dec 2021 – June 2022

Robust autonomous vehicle vision training through adversarial image generation using GANs: generation of road features with StyleGAN3, perturbed image generation training using stop sign dataset scraped from DALL-E (2021) prerelease.

Industry experience

Cisco Systems, Secure Equipment Access

San Jose, California

Software Engineering Intern II

Summer 2024

Extensible Golang asset connection workflow engine for device health and security monitoring using Splunk and Ollama, reduced batch computing overhead through multi-threaded integration in REST API server and SSH client engine.

Cisco Systems, Internet of Things

San Jose, California

Software Engineering Intern I

Summer 2023

IoT troubleshooting dashboard and architecture development: Implemented application manager microservices for accessible troubleshooting platform, Applied Apache Freemarker templating enabling runtime workflow modification, Developed REST API workflow for edge device communication.

Cisco Systems, Webex San Jose, California
Software Engineering Intern I Summer 2022
K8s and Docker DevOps for service cost reduction: Facilitated transition to AWS Spot EC2 and Graviton ARM Instances, Prototyped automatic node shutdown, cluster scaling, and Helm-based app installer.

Posters and Talks

Fair Augmentation of Decision Trees Through Selective Node Retraining
Baylearn Symposium - Poster Presentation Oct 2023

Diffusion Models for Species-Specific Functional Habitat Connectivity
Google Geo For Good Summit - Lightning Talk Oct 2023

Student Leadership

UCSC ACM Chapter, President Sep 2022 – June 2024
UCSC's largest CSE student organization of over 1,000 members: Hosted weekly technical workshops, Co-founded undergraduate AI research lab, Sponsored student-run SWE projects, Managed annual collegiate hackathon.

Santa Cruz Artificial Intelligence, Director of Instruction Sep 2022 – June 2024
UCSC's premier AI/ML student organization: Designed curriculum introducing students to AI/ML methodology, Led weekly application-based workshops, Assisted with quarterly AI project competitions.

Google Developer Student Club, Director of Instruction Sep 2022 – June 2024
UCSC's GDSC chapter for software development: Developed coding workshops for weekly instructional meetings, Taught usage of Google Cloud Platform, TensorFlow, and Kubernetes.

Honors and scholarships

Chancellor's Undergraduate Research Award (UCSC) 2024
Dean's Undergraduate Research Award (UCSC) 2024
Dean's Award (UCSC) 2021 – 2024